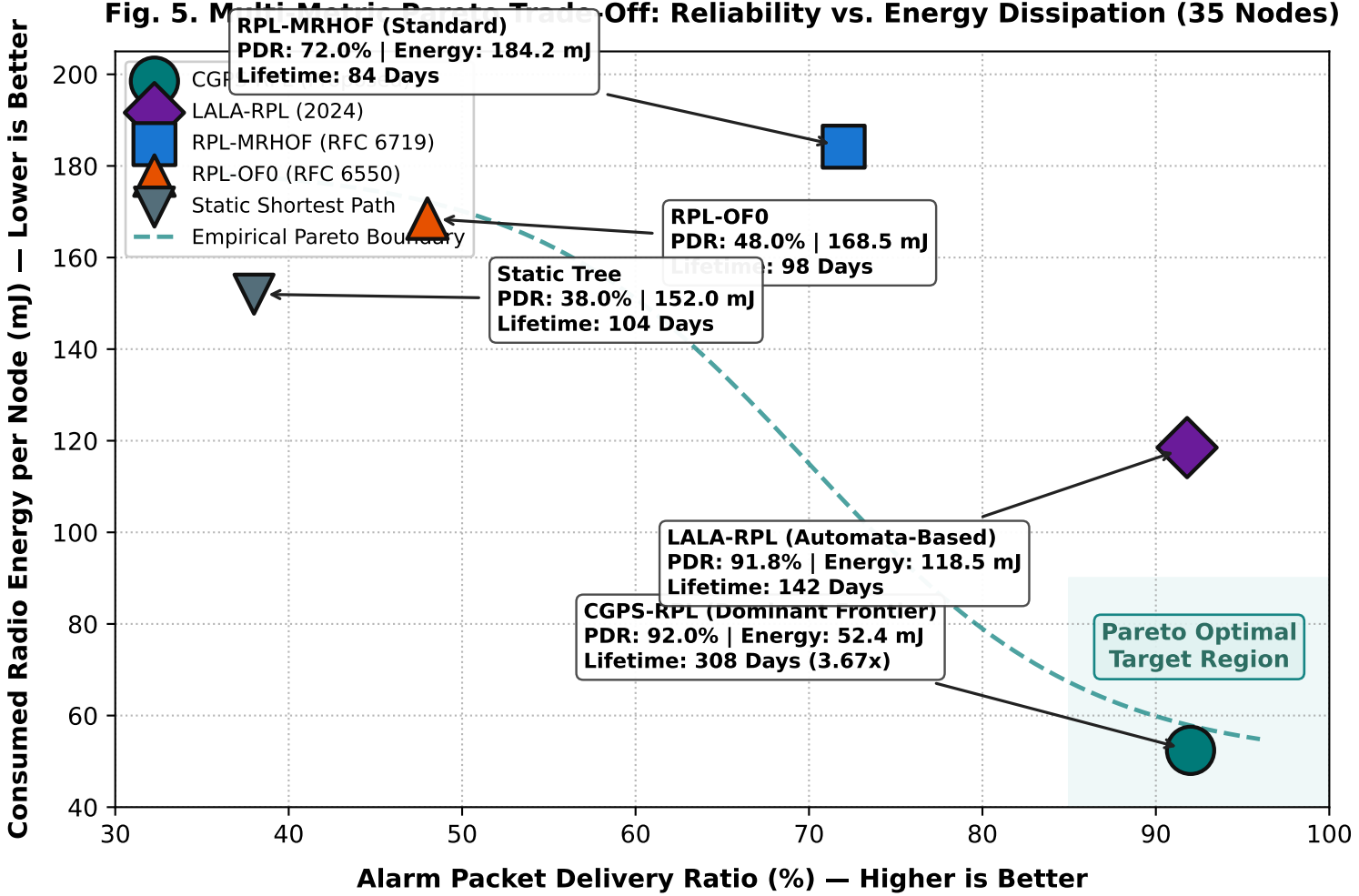
Fig. 5. Multi-Metric Pareto Trade-Off: Reliability vs. Energy Dissipation (35 Nodes)
RPL-MRHOF (Standard)
PDR: 72.0% | Energy: 184.2 mJ
Lifetime: 84 Days
CGP
LALA-RPL (2024)
RPL-MRHOF (RFC 6719)
RPL-OF0 (RFC 6550)
Static Shortest Path
Empirical Pareto Boundary
RPL-OF0
PDR: 48.0% | 168.5 mJ
: 98 Days
Static Tree
PDR: 38.0% | 152.0 mJ
Lifetime: 104 Days
LALA-RPL (Automata-Based)
PDR: 91.8% | Energy: 118.5 mJ
Lifetime: 142 Days
CGPS-RPL (Dominant Frontier)
PDR: 92.0% | Energy: 52.4 mJ
Lifetime: 308 Days (3.67x)
Pareto Optimal
Target Region
Consumed Radio Energy per Node (mJ) — Lower is Better
Alarm Packet Delivery Ratio (%) — Higher is Better
200
180
160
140
120
100
80
60
40
30
40
50
60
70
80
90
100